Training a small image model

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

TEACHER GUIDE · 1 OF 9

Prepare the purpose, prerequisites and materials

> Prepare Training a small image model: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Trace a complete image model with fixed features and one trained weight

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Pooling and spatial information

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

Look for: Compute max and mean pooling under an explicit window rule

Reference example, not the only acceptable wording: Teacher explanation: Values are 1, 3, 2 and 7. Sum is 13, maximum 7 and mean 3.25. Each pooled **output** has shape 2 by 2.

Image tensors and channels

Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

Look for: Read a pixel using declared row column and channel axes

Reference example, not the only acceptable wording: Teacher explanation: The RGB pixel is [0, 0, 255]; its blue value is 255. Shape is 2 by 2 by 3, with four pixels and twelve channel values.

Initialisation and reproducibility

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

Look for: Trace how starting values affect a fixed **training** budget

Reference example, not the only acceptable wording: Teacher explanation: States are -2, 0 and 1. Final **training** MSE is 2.5. Each step halves the signed distance from 2.

For the compact alternative schedule, use two 45-minute sessions. The model deliberately trains only one **output** scalar; do not present the fixed kernels as learned.

Keep the final cases below teacher-only until students record the selected **checkpoint** and threshold. Supply each final image first for **prediction**, then reveal its label.

No image download, account or external **training** library is needed.

Outcomes

Trace a complete image model with fixed features and one trained weight

Select a **checkpoint** using a declared validation rule

Evaluate a frozen model against a **baseline** and limit the conclusion

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

Moving the same values between left and right can preserve the maximum and mean. A summary of all four positions may discard the distinction the task needs.

End goal for the pictorial case

By the end, I can explain **training** a small image model using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain and model the mechanism

> Model Training a small image model using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Training a small image model · 1

Use fictional 2 by 2 binary images, stored as two rows. Predict class 1 when the left column sum exceeds the right column sum; otherwise predict class 0. Equal sums belong to class 0.

This known arithmetic task makes every step checkable; it does not establish real-world image recognition.

Apply the fixed kernel [[1, -1], [1, -1]] to the entire image to obtain d. Its negative gives -d. ReLU produces h1 = max(0, d) and h2 = max(0, -d). The **output** score is a(h1 - h2) = ad.

Only the shared scalar a is trained; both kernels stay frozen and there is no bias or pooling.

For squared-error **training**, map class 1 to target t = 1 and class 0 to t = -1. Predict class 1 only when score > 0; otherwise class 0. The score is not a probability.

Keep the teacher-held final cases unseen until the **checkpoint** is selected and recorded.

Explain Training a small image model · 2

Start a = 0. Use all four **training** examples in each update, ordinary mean squared error and learning rate 0.1. Save **checkpoint**s at updates 0, 1 and 2. The **training** loss is 2.5a squared - 3a + 1, with gradient 5a - 3.

Subtract learning rate times gradient; this gives a = 0.3 then a = 0.45.

Explain Training a small image model · 3

The code assumes valid 2 by 2 numeric images and nonempty labelled lists. It prints update, weight, **training** MSE and validation MSE. Validation examples are scored but never used in the gradient.

A single-channel filter spanning the whole image produces one value here, so no further spatial pooling is needed.

Term	Meaning
Frozen feature	A feature computation whose parameters are kept unchanged during training
Checkpoint	Saved model parameters at a particular training stage
Selection metric	The quantity used under a declared rule to choose a model
Baseline	A simple comparison method evaluated on the same cases

Compare the saved checkpoints

UPDATE AND WEIGHT	TRAINING MSE	VALIDATION MSE
0 with a 0	1	1
1 with a 0.3	0.325	0.49
2 with a 0.45	0.15625	0.3025

Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

Explain Training a small image model · 4

Guided: at a = 0.3, E and F score 0.3 and -0.3; at a = 0.45 they score 0.45 and -0.45. Both **checkpoint**s predict classes 1 and 0 correctly, while squared residuals change from 0.49 to 0.3025.

Final cases: G rows [1, 1], [1, 0], class 1; H rows [1, 1], [0, 1], class 0; I rows [1, 0], [0, 1], class 0. Differences are 1, -1, 0; scores 0.45, -0.45, 0; **prediction**s 1, 0, 0. Model accuracy is 3 of 3, **baseline** 2 of 3.

Signed-target squared errors are 0.3025, 0.3025, 1, so final MSE is 0.535. The tie case is classified correctly despite nonzero regression error.

Exit: only a was learned; using final results to choose a model contaminates final evaluation. Preserve setup, **prediction**s, **baseline** and limits.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the saved checkpoints

UPDATE AND WEIGHT	TRAINING MSE	VALIDATION MSE
0 with a 0	1	1
1 with a 0.3	0.325	0.49
2 with a 0.45	0.15625	0.3025

Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

At a = 0.3 and a = 0.45, compute the validation scores for E and F and their predicted classes. Why can validation accuracy stay the same while validation MSE changes?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Training a small image model

Train one tiny image score with a fixed budget and a separately reserved check.

The facts we will use

Supplied facts

Toy brightness difference d=left−right; score=a×d. **Training** pair d=2, target 2; start a=0; squared loss; learning rate 1/16; two updates.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/23

Training a small image model

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. First update

Gradient 2(2a-2) x 2=-8; new a = 0.5, training loss 1.

2. Second update

Gradient at 0.5 is-4; new a = 0.75, training loss 0.25.

3. Validation score

At reserved d1, target 1: prediction 0.75, squared error 0.0625.

CHANGE ONE THING

Double the two-update budget while keeping the same training rule.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First update	Gradient 2(2a-2)×2=-8; new a = 0.5, training loss 1.
2. Second update	Gradient at 0.5 is-4; new a = 0.75, training loss 0.25.
3. Validation score	At reserved d1, target 1: prediction 0.75, squared error 0.0625.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

What this example does not establish

Keep the stated limit

This scalar brightness model is a transparent toy image task, not a claim that a full image network was run.

Change one thing in the pictured case

Change one thing

Change: Double the two-update budget while keeping the same **training** rule.

Predicted result

Prediction: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625.

Why the result follows

Reason: Each update halves the distance to the **training** optimum 1 in this toy system.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Training a small image model, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the pooling collision and declare the binary image task, signed targets and separate splits.
7 to 18 minutes	Trace fixed kernels, ReLU features and the scalar head, then complete W1.
18 to 32 minutes	Compute both full-data updates and use W2 to check the loss arithmetic.
32 to 45 minutes	Run or trace the code and use W3 to select a saved checkpoint by validation MSE.
Session 2 start to 8 minutes	Complete the guided validation scores; record the frozen model and constant baseline.
8 to 20 minutes	Reveal final images G, H and I only after the record is complete; collect predictions before labels, then compare both methods.
20 to 35 minutes	Collect W4 independently, requiring validation calculations for every checkpoint.
35 to 45 minutes	Use W5 and the exit response to audit claims and administer the fresh retry.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a ledger for image, difference, signed target, score, residual and class. Allow calculators and distinguish class 0 from signed target -1.

Extension: enumerate all sixteen binary 2 by 2 images under the frozen model and exact task rule; label this exhaustive toy-domain audit separately from the three-case held-out evaluation.

Reduce help deliberately

Completion practice

Completion task: Starting at a = 0, use gradient 5a - 3 and rate 0.1 for two updates. Give each gradient, new weight and **training** MSE.

Reasoning to listen for

Teacher check: Teacher explanation: First gradient -3 gives a = 0.3 and MSE 0.325. Second gradient -1.5 gives a = 0.45 and MSE 0.15625. Each gradient is computed at the current weight before the update.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Training** a small image model. Proposed explanation: “All parameters in a trained model must have been learned”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: The declared experiment freezes the feature kernels and adjusts only a.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

No. Record the extra budget and score suitable validation **evidence**; use an untouched final evaluation after selecting a schedule. Changed-case check: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625.

Each update halves the distance to the **training** optimum 1 in this toy system.

Changed-case reasoning: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Each update halves the distance to the **training** optimum 1 in this toy system.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Training a small image model investigation and use the first unsupported step to choose a teaching response.

All parameters in a trained model must have been learned: The declared experiment freezes the feature kernels and adjusts only a.

Perfect accuracy means zero MSE: Thresholded classes can all be correct while continuous scores differ from signed targets.

The last **checkpoint** must be best: Selection depends on the declared validation rule, not update order or **training** loss alone.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Training a small image model questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: at a = 0.3, E and F score 0.3 and -0.3; at a = 0.45 they score 0.45 and -0.45. Both **checkpoint**s predict classes 1 and 0 correctly, while squared residuals change from 0.49 to 0.3025.

Final cases: G rows [1, 1], [1, 0], class 1; H rows [1, 1], [0, 1], class 0; I rows [1, 0], [0, 1], class 0. Differences are 1, -1, 0; scores 0.45, -0.45, 0; **prediction**s 1, 0, 0. Model accuracy is 3 of 3, **baseline** 2 of 3.

Signed-target squared errors are 0.3025, 0.3025, 1, so final MSE is 0.535. The tie case is classified correctly despite nonzero regression error.

Exit: only a was learned; using final results to choose a model contaminates final evaluation. Preserve setup, **prediction**s, **baseline** and limits.

W1 Trace features and a prediction

Workbook question · W1

For Train B, compute d, h1, h2 and the score at a = 0.45. Give the class **prediction** and squared error against its signed target.

Teacher reasoning · W1

Teacher explanation: d = 2, h1 = 2 and h2 = 0. Score is 0.9, **prediction** class 1 and squared error (0.9 - 1) squared = 0.01.

W2 Check both updates

Workbook question · W2

Starting at a = 0, use gradient 5a - 3 and rate 0.1 for two updates. Give each gradient, new weight and **training** MSE.

Teacher reasoning · W2

Teacher explanation: First gradient -3 gives a = 0.3 and MSE 0.325. Second gradient -1.5 gives a = 0.45 and MSE 0.15625. Each gradient is computed at the current weight before the update.

W3 Apply the selection rule

Workbook question · W3

Select among the three saved lesson **checkpoint**s using validation MSE, with earliest update breaking a tie. Is selecting by **training** accuracy the same declared procedure?

Teacher reasoning · W3

Teacher explanation: Select update 2 with a = 0.45 and validation MSE 0.3025.

Training accuracy is a different **selection metric**; both trained **checkpoint**s have perfect **training** accuracy, so it does not establish the same choice.

W4 Choose a checkpoint independently

Workbook question · W4

A separate model scores bd. Its **training** examples have d = 1 and -1 with signed targets 1 and -1. Start b = 0; gradient is 2(b - 1), rate 0.25 and budget two updates. Validation differences are 2 and -2 with targets 1 and -1.

Calculate all three weights and their validation MSEs, then select by lowest validation MSE.

Teacher reasoning · W4

Teacher explanation: Weights are 0, 0.5 and 0.75. Validation scores are [0, 0], [1, -1] and [1.5, -1.5], so validation MSEs are 1, 0 and 0.25. Select update 1 with b = 0.5, even though a later update exists.

Training MSE falls from 1 to 0.25 to 0.0625.

W5 Audit a final evaluation

Workbook question · W5

After inspecting the final cases, a team changes the decision threshold and reports the best result as an unseen test score. Explain the problem and list two things the original report should preserve.

Teacher reasoning · W5

Teacher explanation: Final outcomes influenced the revised model, so that set is no longer unseen **evidence** for the revision.

Preserve the original frozen **checkpoint** and threshold, per-case **prediction**s and errors, **baseline** results, and split IDs. Evaluate the revision on new reserved cases.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires weights 0, 0.5, 0.75 and validation MSEs 1, 0, 0.25; select update 1. Fresh retry: train on d = 2 and -2, signed targets 1 and -1; gradient 8b - 4, rate 0.0625, start 0 and two updates. Weights are 0, 0.25, 0.375.

Validate on d = 4 and -4 with targets 1 and -1: MSEs are 1, 0, 0.25. Select update 1 with b = 0.25.

Assess independent transfer

Independent question · withhold the key

Selected case: A separate model scores bd. Its **training** examples have d = 1 and -1 with signed targets 1 and -1. Start b = 0; gradient is 2(b - 1), rate 0.25 and budget two updates.

Validation differences are 2 and -2 with targets 1 and -1. Calculate all three weights and their validation MSEs, then select by lowest validation MSE.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: Weights are 0, 0.5 and 0.75. Validation scores are [0, 0], [1, -1] and [1.5, -1.5], so validation MSEs are 1, 0 and 0.25. Select update 1 with b = 0.5, even though a later update exists.

Training MSE falls from 1 to 0.25 to 0.0625.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Training a small image model to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled **training** a small image model **evidence** sheet to Image-pattern Network Investigation. Show how you can trace a complete image model with fixed features and one trained weight.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a ledger for image, difference, signed target, score, residual and class. Allow calculators and distinguish class 0 from signed target -1.

Extension: enumerate all sixteen binary 2 by 2 images under the frozen model and exact task rule; label this exhaustive toy-domain audit separately from the three-case held-out evaluation.

Week 24 examines augmentation and leakage, including how related image variants should remain in the same split.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/24, ai-5/26, ai-5/29. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Training a small image model lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Trace a complete image model with fixed features and one trained weight

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Starting at a = 0, use gradient 5a - 3 and rate 0.1 for two updates. Give each gradient, new weight and **training** MSE.

Supported practice answer

Reasoned practice answer: Teacher explanation: First gradient -3 gives a = 0.3 and MSE 0.325. Second gradient -1.5 gives a = 0.45 and MSE 0.15625. Each gradient is computed at the current weight before the update.

Check every learner before discussion: Use the worked case for **Training** a small image model. Proposed explanation: “All parameters in a trained model must have been learned”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: The declared experiment freezes the feature kernels and adjusts only a.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Double the two-update budget while keeping the same **training** rule. Require a **prediction** and a reason before comparing. Expected result: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625.

Reason: Each update halves the distance to the **training** optimum 1 in this toy system.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project **checkpoint**: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

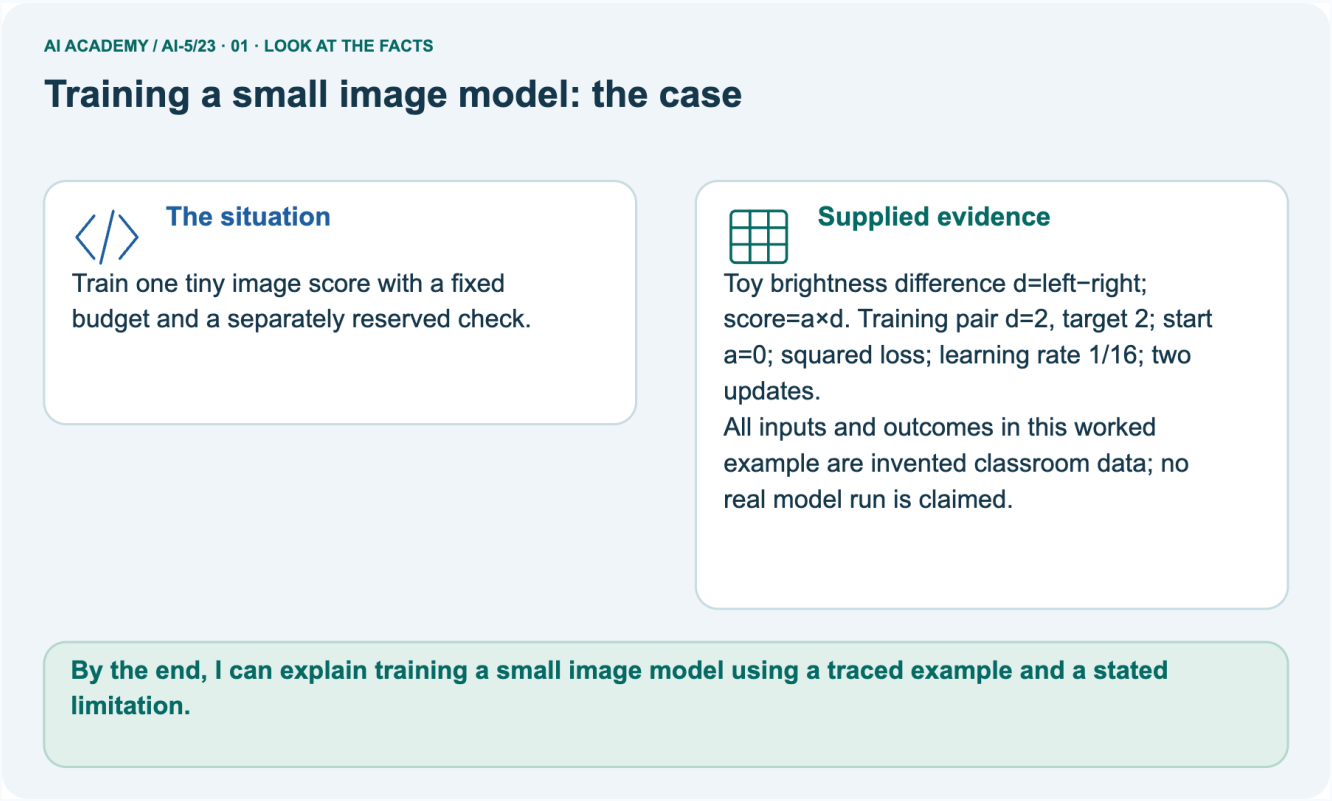
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Training a small image model graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Training a small image model: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Train one tiny image score with a fixed budget and a separately reserved check. Supplied facts: Toy brightness difference d=left−right; score=a×d.

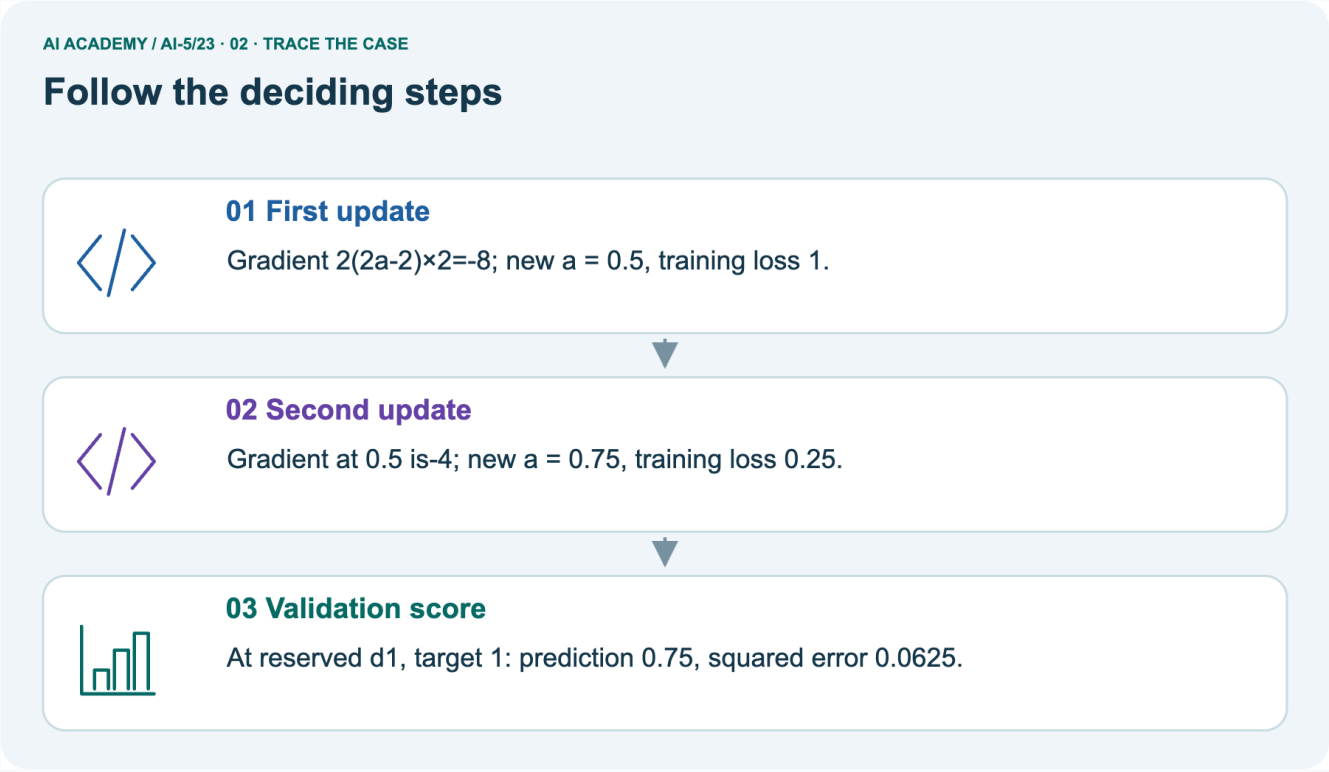
Training pair d=2, target 2; start a=0; squared loss; learning rate 1/16; two updates. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

First update: Gradient 2(2a-2)×2=-8; new a = 0.5, **training** loss 1. → Second update: Gradient at 0.5 is-4; new a = 0.75, **training** loss 0.25. → Validation score: At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Compare the saved checkpoints

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Compare the saved checkpoints

Update and weight	Training MSE	Validation MSE
0 with a 0	1	1
1 with a 0.3	0.325	0.49
2 with a 0.45	0.15625	0.3025

Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

Follow the labelled picture with the matching explanation. Open the vector diagram

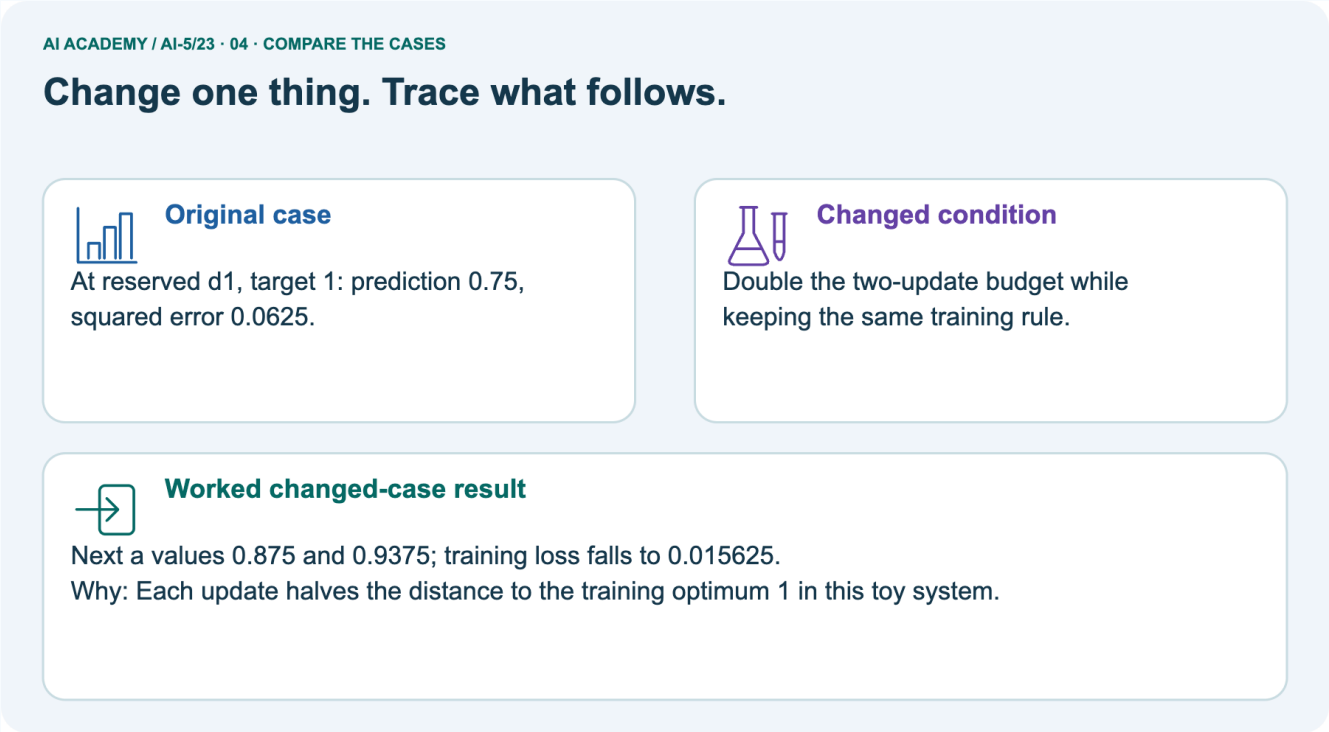
Compare the saved **checkpoint**s. 0 with a 0 | 1 | 1. 1 with a 0.3 | 0.325 | 0.49. 2 with a 0.45 | 0.15625 | 0.3025

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At reserved d1, target 1: **prediction** 0.75, squared error 0.0625. Changed condition: Double the two-update budget while keeping the same **training** rule.

Expected result: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Reason: Each update halves the distance to the **training** optimum 1 in this toy system.

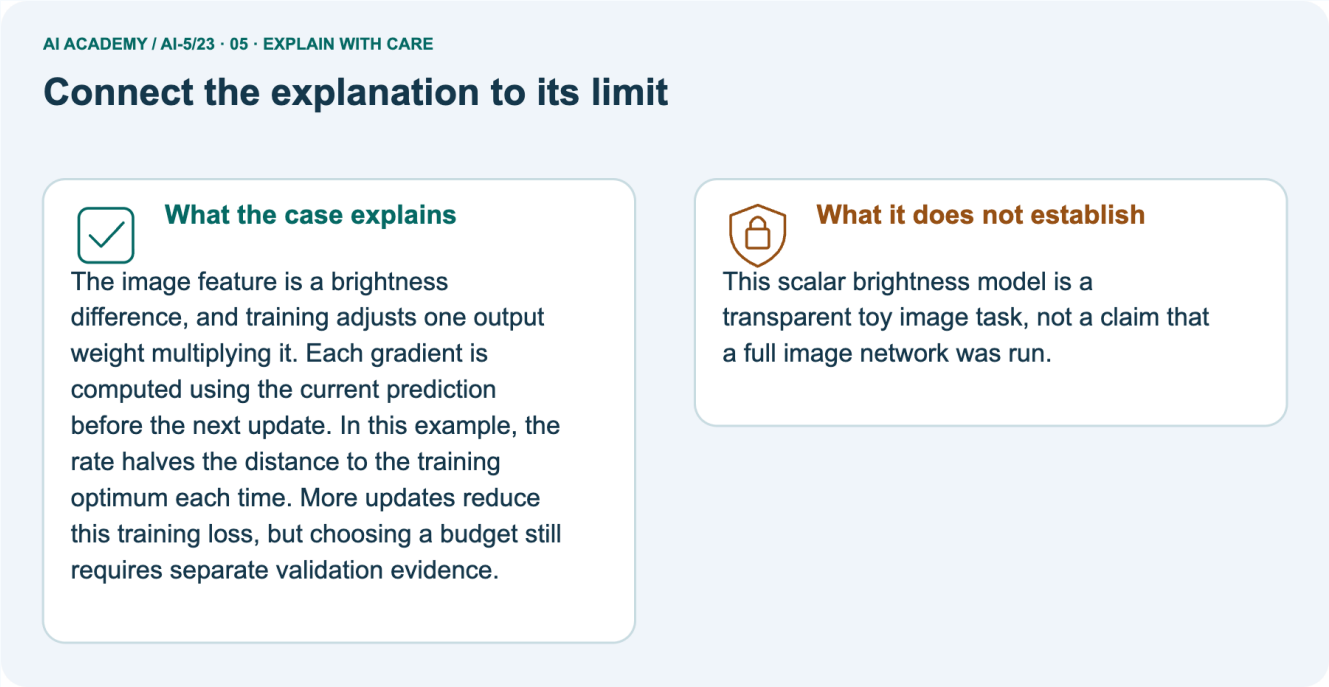
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Each update halves the distance to the **training** optimum 1 in this toy system.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

Limit: This scalar brightness model is a transparent toy image task, not a claim that a full image network was run.

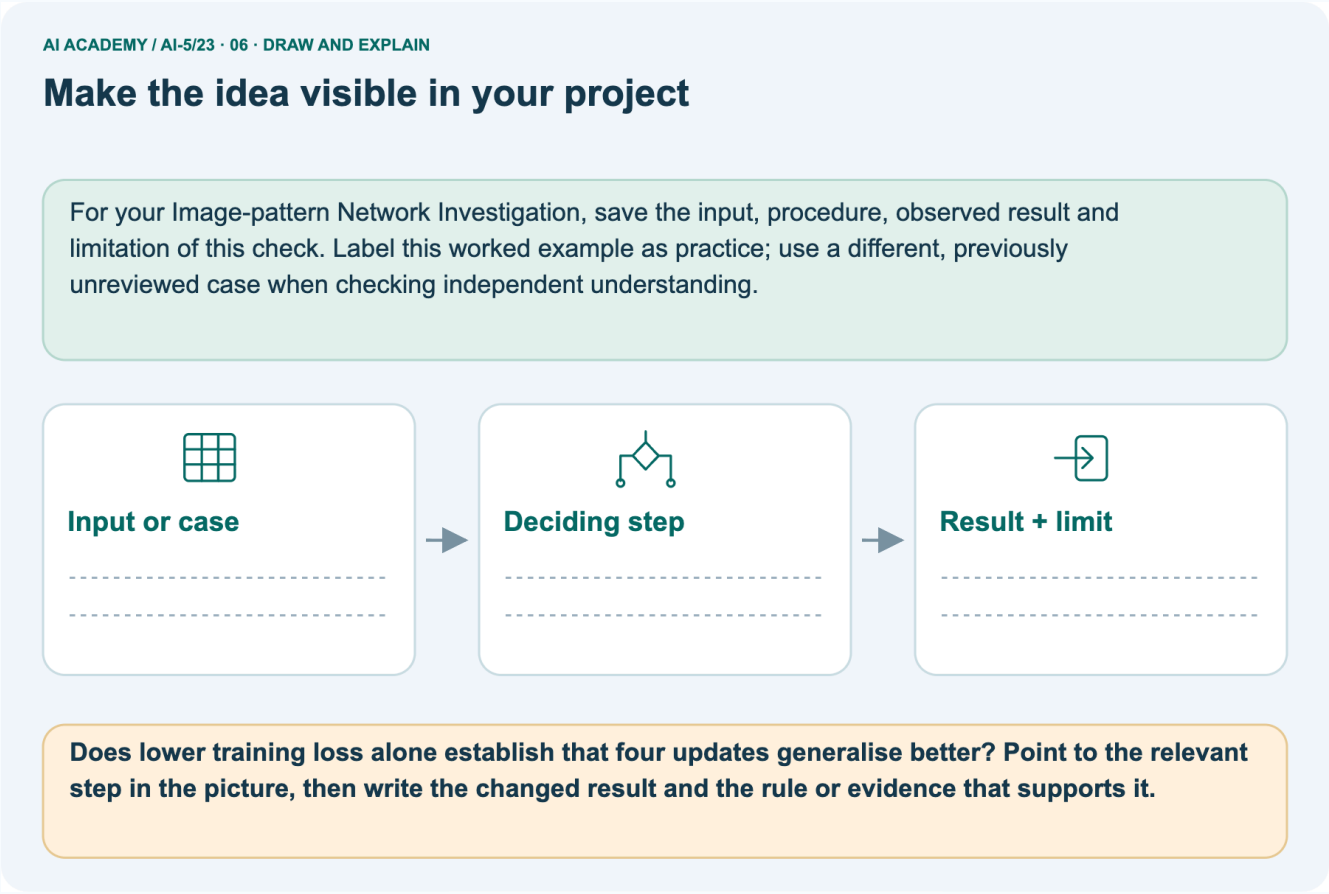
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This scalar brightness model is a transparent toy image task, not a claim that a full image network was run.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. Record the extra budget and score suitable validation **evidence**; use an untouched final evaluation after selecting a schedule.

Changed-case check: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Each update halves the distance to the **training** optimum 1 in this toy system.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.